

Total No. of Questions : 8]

SEAT No. :

PA-1454

[Total No. of Pages : 2

[5926]-70

T.E. (Computer Engineering)

CLOUD COMPUTING

(2019 Pattern) (Semester - II) (Elective-II) (310254C)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary

Q1) a) Describe CPU, Network and Storage Virtualization? [9]

b) Draw and Explain the Virtualization Architecture in detail? [8]

OR

Q2) a) Define Virtualization? Explain different types of Virtualizations? [9]

b) Differentiate between Virtualization in Grid and Virtualization in Cloud [8]

Q3) a) What are the components of Microsoft Azure? Explain briefly? [9]

b) How cloud computing can be used for business and consumer applications like ERP or CRM? [8]

OR

Q4) a) Describe the Amazon Database Services? [8]

b) Explain Google Cloud Applications in detail? [9]

Q5) a) What are the security issues of cloud computing identified by cloud security alliance (CSA)? Explain any three in detail? [9]

b) How Trusted Cloud Computing can be used to manage the risk and security in a cloud? [9]

OR

P.T.O.

- Q6) a)** Explain the six step risk management processes? [9]
- b)** Describe how to perform Secure Cloud Software Testing? [9]

- Q7) a)** What do you mean by IoT Cloud? And how IoT cloud can be used in home automation? [9]
- b)** Explain architecture and working of Kubernetes. [9]

OR

- Q8) a)** What are the future trends in cloud computing? Explain in brief? [9]
- b)** Differentiate between Distributed Cloud Computing Vs Edge Computing? [9]



Total No. of Questions: 8]

SEAT No. :

P280

[6003]-359

[Total No. of Pages : 2

T.E. (Computer Engineering/A.I.D.S.)

CLOUD COMPUTING

(2019 Pattern) (Semester-II) (Elective II) (310254 C)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Differentiate between grid computing and cloud computing? **[9]**

b) Define virtualizations? Explain the advantages and disadvantages of Virtualization? **[8]**

OR

Q2) a) Describe virtual clustering in cloud computing? **[9]**

b) Explain the importance of hypervisor in cloud computing? Compare Type 1 and Type 2 hypervisor? **[8]**

Q3) a) Enlist an applications of cloud computing in differnt Area? Describe any two applications? **[9]**

b) Explain the different components of AWS? **[8]**

OR

Q4) a) How the Amazon simple storage service (S3) works? Explain with suitable diagram? **[8]**

b) Enlist the steps for configuring Amazon EC2 VM instance? **[9]**

Q5) a) What are the different types of testing in cloud computing? Explain briefly? **[9]**

b) Explain the different types of security risk involved in cloud computing? **[9]**

OR

P.T.O.

- Q6)** a) Describe the different Cloud Security Services in detail? [9]
b) State the use of Content Level Security (CLS)? [9]

- Q7)** a) Describe client-server architecture of docker? [9]
b) Explain Mobile Cloud in detail? [9]

OR

- Q8)** a) Differentiate between Distributed Cloud Computing Vs Edge Computing? [9]
b) Explain the concept of DevOps in detail? [9]



Total No. of Questions : 8]

SEAT No. :

P-7550

[Total No. Of Pages : 2

[6180]-58

T.E. (Computer/A.I & D.S)

CLOUD COMPUTING

(2019 Pattern) (Semester - II) (310254 C) (Elective - II)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q. 1 or Q. 2, Q. 3 or Q. 4, Q. 5 or Q. 6, Q. 7 or Q. 8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Explain virtual clustering in cloud computing? [9]

b) Describe the importance of hypervisor in cloud computing? Differentiate Type I and Type 2 hypervisor? [8]

OR

Q2) a) Compare grid computing and cloud computing? [9]

b) Define virtualizations? Describe the advantages and disadvantages of Virtualization? [8]

Q3) a) Draw & explain the working of Amazon simple storage service(S3)? [9]

b) Describe the different steps for configuring Amazon EC2 VM instance?[8]

OR

Q4) a) Enlist an applications of cloud computing in different Area? Describe any two applications? [8]

b) Identify the different components of AWS? [9]

P.T.O.

- Q5) a)** Explain the different Cloud Security Services in detail? [9]
b) State the use of Content Level Security (CLS)? [9]

OR

- Q6) a)** What are the different types of testing in cloud computing? Explain briefly? [9]
b) Analyze the different types of security risk involved in cloud computing? [9]
- Q7) a)** Draw & Explain client - server architecture of docker? [9]
b) Explain Multimedia Cloud in detail? [9]

OR

- Q8) a)** Differentiate between Distributed Cloud Computing Vs Edge Computing? [9]
b) Describe the concept of DevOps in detail? [9]



Total No. of Questions : 8]

SEAT No. :

PB-3789

[Total No. of Pages : 2

[6262]-48

**T.E. (Computer / Artificial Intelligence and Data Science
Engineering)**

CLOUD COMPUTING

(2019 Pattern) (Semester - II) (310254C) (Elective-II)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4 Q.5 or Q.6, Q. 7or Q.8.*
- 2) *Figures to the right indicate full marks.*
- 3) *Neat diagram must be drawn wherever necessary.*
- 4) *Assume suitable data if necessary.*

- Q1)** a) What is virtualization? What is Type 1 Hypervisor and Type 2 Hypervisor? [6]
- b) Explain Virtual clustering in detail? [6]
- c) Explain Virtualization in grid computing? [6]

OR

- Q2)** a) Explain Virtualization Application and Pitfalls of Virtualization? [6]
- b) Explain Network and Storage Virtualization? [6]
- c) Explain virtual machine migration technique in detail? [6]

- Q3)** a) What is AWS? What are the services provided by AWS? [6]
- b) Explain amazon S3 and Amazon EC2? [6]
- c) Explain SQL Azure in detail? [5]

OR

- Q4)** a) Explain Google App Engine with its installation steps? [6]
- b) Draw and explain Architecture of Amazon Dynamo? [6]
- c) Differentiate between Dynamo DB and Amazon S3? [5]

P.T.O.

- Q5)** a) What is role of Confidentiality, Integrity and Availability in Cloud Computing? [6]
b) Explain types of Risks in Cloud Computing? [6]
c) Explain the secure cloud software testing? [6]

OR

- Q6)** a) Explain the cloud security services in details? [6]
b) Write a short note on content level security? [6]
c) Compare server side and client-side encryption? [6]

- Q7)** a) Explain the mobile cloud computing? [6]
b) Explain docker with its Architecture? [6]
c) Explain the application of IOT and cloud in your home? [5]

OR

- Q8)** a) What is Energy aware cloud computing? Explain in details? [6]
b) Explain container & Kubernetes in detail? [6]
c) Explain Distributed cloud computing? [5]



Total No. of Questions : 8]

SEAT No. :

PC1731

[Total No. of Pages : 2

[6353]-48

T.E (Computer Engg./Artificial Intelligence and Data Science Engg.)

CLOUD COMPUTING

(2019 Pattern) (Semester - II) (Elective-II) (310254(C))

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6 and Q7 or Q8.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicates full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Define Virtualization? Describe different types of Virtualizations? [9]

b) Differentiate between Virtualization in Grid and Virtualization in Cloud? [8]

OR

Q2) a) Describe CPU, Network and Storage Virtualization? [9]

b) Draw and Explain the Virtualization Architecture in detail? [8]

Q3) a) Describe the Amazon Database Services? [9]

b) Explain Google Cloud Application in detail? [8]

OR

Q4) a) Identify the different components of Microsoft Azure? explain Briefly? [8]

b) How cloud computing can be used for business and consumer applications like ERP or CRM? [9]

Q5) a) What are the security issues of cloud computing identified by cloud security alliance (CSA)? explain any three in detail? [9]

b) How Trusted Cloud Computing can be used to manage the risk and security in a cloud? [9]

OR

Q6) a) Describe the six step risk management processes? [9]

b) Describe how to perform Secure Cloud Software Testing? [9]

P.T.O.

- Q7)** a) What do you mean by IoT Cloud? How IoT cloud can be used in home automation? [9]
b) Draw & Explain architecture of Kubernetes. [9]

OR

- Q8)** a) Identify the future trends in cloud computing? Explain in brief? [9]
b) Differentiate between Distributed Cloud Computing Vs Edge Computing? [9]



Total No. of Questions : 8]

SEAT No. :

PD4253

[Total No. of Pages : 2

[6403]-48

T.E. (Artificial Intelligence and Data Science) (Computer Engg.)

CLOUD COMPUTING

(2019 Pattern) (Semester - VI) (310254 C) (Elective - II)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *Figures to the right indicate full marks.*
- 3) *Neat diagram must be drawn wherever necessary.*
- 4) *Assume suitable data if necessary.*

- Q1)** a) Define virtualization. Discuss the types of virtualizations. [6]
- b) Draw & Explain VMware architecture in detail. [6]
- c) Distinguish between virtualization in Grid and virtualization in Cloud. [6]

OR

- Q2)** a) Explain the role of virtualization in cloud computing. [6]
- b) Discuss how distributed computing technologies underpin the architecture of cloud services, focusing on the role of load balancing and fault tolerance? [6]
- c) Draw & Explain the anatomy of cloud infrastructure. [6]
- Q3)** a) List and explain the services offered by AWS. [6]
- b) Explain the AWS Elastic load Balancer and its types with its advantages. [6]
- c) Draw and Explain Architecture of Amazon Dynamo. [5]

OR

P.T.O.

- Q4)** a) Explain steps to configure server for EC2. [6]
b) Describe the storage Services offered by Amazon. [6]
c) Write a note on Services offered by Microsoft. [5]

- Q5)** a) Identify and explain Security issues in Cloud Computing. [6]
b) Discuss Secure Cloud Software Requirements in detail. [6]
c) List and explain Security Authorization Challenges in the Cloud. [6]

OR

- Q6)** a) Explain the Secure Cloud Software Testing. [6]
b) Discuss Challenges, Advantages and Disadvantages of Data Security in Cloud. [6]
c) Explain Cloud Digital persona and Data security. [6]

- Q7)** a) Difference between Distributed Cloud Computing Vs Edge Computing. [6]
b) Describe DevOps Architecture in Cloud Computing. [6]
c) Explain the Comet Cloud Architecture with diagram. [5]

OR

- Q8)** a) Explain Containers and Dockers in details. [6]
b) Describe Jungle Computing in detail. [6]
c) Draw & explain working of Kubernetes architecture. [5]

